$$\vec{u}_g^{ac}(t)=\left[u_u(t)\quad u_v(t)\quad u_w(t)\right]^T$$

$$\vec{i}_g^{ac}(t)=\left[i_u(t)\quad i_v(t)\quad i_w(t)\right]^T$$

$$u_g^{dc}(t)$$

$$i_g^{dc}(t)$$

$$\rightarrow \quad \min J\left[\vec{u}_{\xi\eta}^{p}(t),\vec{u}_{\xi\eta}^{n}(t),\vec{i}_{\xi\eta}^{p}(t),\vec{i}_{\xi\eta}^{n}(t),u_g^{dc}\right] \quad \rightarrow \quad \begin{cases}\vec{i}_{\xi\eta}^{p}\big|^{ref}(t)\\ \vec{i}_{\xi\eta}^{n}\big|^{ref}(t)\end{cases}$$

Contributors from renewable energy sources

